

Abstract—Abstract goes here.

Index Terms—COPD, wearable device, respiratory monitoring, human-robot interaction

I. INTRODUCTION

Chronic obstructive pulmonary disease (COPD) is a progressive chronic lung disease associated with persistent respiratory symptoms, such as breathlessness [1]. Activity related breathlessness is one of the most distressing and problematic symptoms experienced by people suffering from COPD, and is closely associated with reduced capacity for exercise and physical inactivity [2], [3] - an outcome that is of particular concern, as it has been shown in several studies that exercise can help alleviate the progression of COPD. Pulmonary rehabilitation (PR), incorporating exercise training and other patient-tailored interventions, can improve dyspnoea (breathlessness) related quality of life and exercise capacity in people with COPD [1].

Despite the impact of breathlessness on the quality of life of COPD patients, it is inherently subjective and thus difficult to quantifiably measure using physiological signals [2], [3]. Despite this, in recent studies, signals such as respiratory rate, heart rate and respiratory muscle activity have demonstrated associations with patient reported dyspnoea and may enable the estimation of breathlessness severity [4]. Wearable monitoring and telemonitoring technologies are also increasingly being studied within PR to collect physiological and symptom data, and to support personalised feedback [5].

II. LITERATURE REVIEW

Breathlessness is both subjective, and multidimensional in how patients experience it, and as such, severity is generally assessed through the self-reporting of patients, rather than through physiological measurements alone [2], [3]. Assessment of dyspnoea is done with a variety of methods: the mMRC scale describes limitation to function in daily life, while the Borg CR10 scale is often used to describe and quantify temporary breathlessness intensity during exercise [2]. Because the degree of breathlessness be altered by exertion, comparison of the intensity of symptoms is most useful when ratings are obtained at set points during a standardised exercise stimulus. Simultaneous physiological measurements can then be related to patient reported ratings [2], [3]. Despite the use of self-reporting, physiological signals still contain information that is associated with the degree of perceived dyspnoea. Dörssers et al. synchronised modified-Borg ratings with respiratory-muscle surface electromyography (EMG), respiratory rate (RR), heart rate (HR), oxygen saturation (SpO_2) and activity type during walking and cycling in 28 people with COPD [4]. A model containing RR, HR, activity type and SpO_2 gave fitted Borg

values within one point for 85.97 % of observations, increasing to 87.86 % after the inclusion of parasternal EMG. RR, HR and EMG were significant predictors to breathlessness intensity, while SpO_2 was found to not be independently significant. However, these findings are describing fitted associations, instead of externally validated and subject independent predictions. Additionally, high quality EMG was only available in 40 % of exercise segments. As such, the study supports the physiological estimation of breathlessness, but as of yet, not the reliable detection of breathlessness from a wearable device.

The feasibility of taking relevant measurements from the wrist has been investigated in other studies. Guber et al. compared wrist-based pulse oximetry with a more conventional fingertip reference in healthy participants with COPD [6]. Across 1,138 paired SpO_2 measurements, the wrist device achieved amplitude readings within 3 % of the reference values. However, measurements that were unreliable or collected during motion were excluded from analysis, and breathlessness was not assessed. As such, evidence that physiological signals can be related to breathlessness and evidence that those signals can be measured accurately at the wrist remain separate.

Existing rehabilitation technologies demonstrate that sensing can be integrated with personalised feedback and remote supervision. COPDTrainer used therapist-derived movement parameters and smartphone inertial sensing to give real time feedback on exercise performance during unsupervised COPD rehabilitation [7]. Carvalho et al. combined cardiorespiratory monitoring with breathing phase feedback and physiotherapist supervision in ten people with COPD, going to show the feasibility of integrating physiological sensing and task-specific guidance within pulmonary rehabilitation [8]. Otto-Yáñez et al. similarly demonstrated an eight-week supervised telerehabilitation programme incorporating repeated Borg assessment, wearable monitoring, symptom reporting and predefined recovery procedures/exercises [5]. However, only nine of the fifteen enrolled participants took the study to completion, and real time HR and SpO_2 monitoring during exercise was provided using primarily an O_2 ring, instead of a smartwatch.

Collectively, the literature supports patient reported assessment of exercise related breathlessness, the potential value of physiological and activity signals when correlated with dyspnoea, and the technical feasibility of wearable supported pulmonary rehabilitation. In combination with the clear problem of breathlessness experienced by COPD patients, the necessity for physician-prescribed exercises and the requirement to be able to complete these exercises at a moments notice - at home, in clinical settings, during exercise - creates a compelling argument to create and investigate new portable and effective methods of delivering these exercises and assisting users

through them.

III. CONCEPTUAL DESIGN

This section draws on the findings of the literature review, along with an interview with a local respiratory physiotherapist, to inform the key functions and form of a proposed wearable device to support COPD patients. It covers the key design requirements, the proposed concepts and the justification for the chosen concept, and a high-level overview of the intended system operation.

A. Concept Overview

The working concept is a wrist device with automatic prompting that guides a person with COPD through a breathing exercise during recovery after physical activity. Buttons allow the user to request assistance, respond to prompts and control the session, while haptic cues communicate the breathing routine. Wrist PPG and motion sensing will provide candidate physiological measurements and information about signal quality.

B. Industry Consultation

We interviewed a respiratory physiotherapist based in Manukau, Auckland, with experience treating patients with COPD and related respiratory illnesses. The interview aimed to establish whether the proposed design was feasible and likely to provide real value to people with COPD, and to learn about current state-of-the-art clinical practice in this area. The key findings are summarised below.

The physiotherapist confirmed that breathlessness is a key problem affecting quality of life for people with COPD. With a limited number of clinicians available, a take-home wearable could relieve some of the burden on the healthcare system and on respiratory physiotherapists. The device would be especially useful for newly diagnosed patients learning to self-manage their symptoms. By providing objective data about breathing status, it could also alleviate breathlessness-related anxiety, which can otherwise worsen symptoms. This validates the device's value for both patients and clinicians.

For patients with severe COPD, oxygen saturation is a useful metric that often correlates with the degree of breathlessness. However, it is less important to monitor in mild to moderate cases.

Day-to-day breathlessness management differs between patients. Strategies include adopting different body positions, performing specific breathing exercises, and other techniques to slow breathing, such as blowing cool air from a fan onto the face. Patients often need to try several tactics before finding the one that works best for them. To report breathlessness, patients commonly use the Borg scale, a simple 0–10 rating that requires no health literacy.

In terms of device function, the physiotherapist recommended a clear indicator of the patient's breathlessness status, including a way to signal when medical assistance is needed. Integration with clinicians must also be considered, including ensuring data

storage complies with regulations and handling cases where a patient stops seeing a clinician but continues to use the device.

These key findings were used to inform the design requirements, concept selection and system operation. Based on this discussion, the device will prioritise people with mild to moderate COPD. As clinician involvement remains important in each patient's COPD management, the device is intended as a supplementary tool rather than a replacement for clinicians, and is expected to be especially useful for newly diagnosed and health-anxious patients. The device is not recommended for severe COPD, where clinicians should be deferred to so that treatment is appropriate and tailored to the individual. Although SpO₂ is less critical for mild to moderate COPD, the wrist PPG sensor can estimate it alongside heart rate and respiratory rate, so SpO₂ monitoring is included as a supplementary metric that can be reviewed by clinicians. It is also crucial that the design is intuitive, requiring little to no digital literacy to use.

C. Design Requirements

The proposed device is intended to coach a person with COPD through a breathing exercise after physical activity or breathlessness. The suitable concept must meet the below requirements, with further detail and performance targets listed in Appendix A.

Under user interaction, the user must be able to start, pause and stop a session with minimal interaction, always. The device must also provide clear cues for the actions and timing of the breathing exercise. Finally, the user must be able to report breathlessness and other adverse symptoms easily.

For Monitoring & Reliability, the device should estimate and monitor the user's heart rate, respiratory rate and SpO₂ when the available signals are sufficiently reliable. It must identify times where movement and other interference create unreliable data and avoid using that data for feedback and monitoring. Finally, the device should prompt an intervention if it detects breathlessness, or other concerning user states. See further details in Appendix A.

On top of these requirements, the device must also be comfortable to wear during exercise and long periods. It must also be battery powered, so users can go about their day. All the data must be stored locally on the device and require explicit consent from the user to share with clinicians.

D. Concept Selection and Rationale

The five concepts compared in Appendix B are a wrist-worn device providing manual timed guidance (C1), a wrist-worn device adding physiological monitoring (C2), a wrist-worn device adding automatic prompting (C3), a chest band with the same functionality of C2, and C3 with the addition of a pet robot frog as a physical coach.

We selected C3 because it combines monitoring with prompts to check symptoms and retains manual control. It has the added benefit over C1 and C2 of responding to the user without prompting, instead of needing user input for health benefits. This makes it a better health aid for users with COPD. We discounted C4 because of the reduced comfort and ease of

user interaction daily, although a chest band would see more accurate measurements. The addition of a coach robot frog, which would mimic the breathing exercises for the user to follow along with would be novel, but has an unproven benefit to user health, and further hampers the day-to-day use of the device.

E. System Architecture and Operation

The system performs six functions: acquiring signals, assessing measurements, responding to assessments, reacting to user input, delivering breathing cues and storing relevant data locally. PPG provides physiological sensing, while buttons and haptic feedback provide human–robot interaction through user input and device response.

Signal acquisition: The PPG sensor records an optical signal from the wrist for estimating heart rate, respiratory rate and SpO₂. The motion sensor records wrist movement to provide the user’s activity for the context of the data. Appropriate filtering is applied to clean the signal, which is calibrated for each user’s baseline during rest.

Measurement Assessment: The signals are then checked for interference, supplemented by the motion data. Unreliable measurements are discarded. When the user is resting and little motion interference is detected, heart rate and respiratory rate are used to numerically classify the patient’s breathlessness on the Borg scale.

Assessment Response: The device uses these assessments to determine the feedback required. If the user records prolonged high activity based on IMU measurements, or breathlessness from PPG readings while at rest, the device will prompt the user to rest and begin a breathing exercise. If the user’s heart rate or respiratory rate is concerning high, based on the resting rate of the user, the device will prompt the user to rest and start a breathing exercise. Prompts request user confirmation before guidance starts, with the user able to dismiss them or request assistance independently.

UI Reaction: The device will also be constantly ready to respond to user interaction, allowing a user to start, stop and pause a breathing exercise, along with logging concerning symptoms like breathlessness, and score their symptoms regularly. The device will use this interaction to inform which exercise to guide the user through and for how long. The symptom scale is based on the Borg CR10 scale described in the literature review [2]: the user adjusts a 0–10 breathlessness rating with two buttons, receives vibration feedback for the selected value, and holds both buttons to confirm it.

Delivering Breathing Cues: The device stores several predefined breathing routines and will propose exercises which can be overruled by the user. It delivers haptic cues for the chosen routine’s inhale, exhale and, where applicable, hold phases; pursed-lip breathing is the proposed default after activity. During still periods, wrist PPG may allow the device to estimate breathing phases and compare their timing with the cues, while motion sensing flags unreliable data. The user can pause or stop at any time.

Data Storage: The device stores the PPG and motion data recorded during each breathing session locally. The data would only be able to be exported to a clinician or other medical professionals through a manual process requiring the user’s explicit consent.

Operating sequence: The user can request assistance after activity or respond to an automatic prompt. Once resting, they report their breathlessness and confirm whether to begin. The device delivers breathing cues and monitors usable signals while the user retains start, pause and stop control. If signals become unreliable, measurement-based feedback and automatic assessment are suspended; user-controlled timed guidance remains available. At completion or early stopping, the user is asked to report symptoms again and the session is stored locally.

IV. DETAILED DESIGN

This section presents the detailed mechatronic design of the proposed wearable device, building on the conceptual design outlined in Section III. The design is structured around several core aspects: the mechanical design of the wearable enclosure and its interface with the wrist, the electrical schematic and sensor circuitry required for physiological signal acquisition, and the underlying software/control logic governing the device’s operation. Each of these aspects is discussed in turn, with the accompanying design rationale grounded in the requirements and constraints established in the preceding sections.

A. Mechanical Design

The wearable adopts a standard wristwatch form factor, comprising a single moulded housing and an adjustable strap to accommodate varying wrist sizes. As feedback is delivered exclusively through haptic actuation rather than a visual display, the design omits a watch face, simplifying the enclosure to a single sealed electronics compartment. TPU is used here, being a flexible, durable material commonly used in wearable enclosures. At the area of skin contact housing the PPG sensor, a clear polycarbonate window is integrated to permit optical transmission while protecting the underlying sensor from sweat and mechanical wear. Given the intended use during physical exercise, the device targets an IP67 ingress protection rating to ensure resistance to sweat and moisture. To achieve this while minimising assembly complexity, the housing is to be joined via ultrasonic welding, creating a continuous, seam-free seal without requiring additional sealing components such as O-rings or gaskets. The polycarbonate window is bonded to the housing using a UV-cured optical adhesive. Overall, the mechanical design follows established conventions for wrist-worn wearable devices, prioritising manufacturability and a proven form factor for user comfort. Fig. 1 shows an exploded view of the housing assembly, identifying the key mechanical and sealing components.

B. Electrical and Sensor Circuitry

The electronics comprise three subsystems: physiological sensing, processing and communication, and wireless power

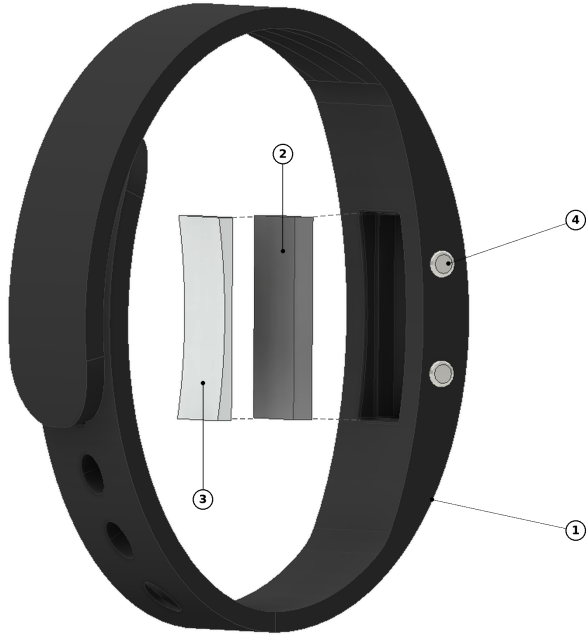


Fig. 1. Exploded view of the wearable housing assembly: (1) TPU housing with integrated watch strap, (2) electronics envelope (simplified), (3) polycarbonate optical window, (4) sealed push buttons.

management, as shown in the full schematic in Fig. 4 in the Appendix. Photoplethysmography (PPG) is acquired using the MAXM86161 optical module [1], which integrates green, red and infrared LEDs with a photodiode and 19-bit ADC in a 2.9mm x 4.3mm x 1.4mm package. The green channel was selected for heart rate (HR) estimation because it more robustly tracks pulsatile flow at the wrist during movement, whereas the red/IR pair enables SpO₂ estimation. Because motion artefact is the dominant source of error in wrist PPG, a BMI270 six-axis IMU [2] provides a synchronous acceleration reference for adaptive artefact cancellation and signal-quality gating; its on-chip activity detection also supports gait and exertion metrics relevant to COPD functional assessment. Both sensors communicate with an ESP32-C3 microcontroller [3] over I²C at 100 Hz, which runs HR/SpO₂ and processes the data locally.

Power is delivered wirelessly through a WR221230-36M8-G receiver coil [4] and BQ51003 Qi receiver [5], feeding a BQ25120A charger [5] that charges a 240 mAh Li-ion cell at 380 mA and supplies a 1.8 V rail for the sensors. A TPS62743 buck converter [6] provides the 3.3 V rail for the MCU. A TPS61240YFFT boost converter [7] provides 5.0 V rail for the green LEDs. Wireless charging was chosen because it removes the need for an exposed connector, allowing a sealed, easily cleaned enclosure, and avoids fine-motor connector handling for older users with reduced dexterity. The estimated average current of 30 mA gives a battery life of approximately 8 h.

Following the TROIKA framework [9], PPG and IMU signals are sampled at 25 Hz. With two sessions per day capped at 30 minutes, each session produces 1.05 MB (23 bytes/sample plus per-second timestamp and CRC), giving 2.1 MB/day

and 63 MB per 30-day review cycle. A 1 Gbit (128 MB) SPI NAND flash Winbond W25N01GV [9] provides a full month of storage with around 2× margin for additional episodes and bad-block reserve.

C. Software and Control Logic

The device's operation is governed by a finite state machine (FSM), shown in full in Fig. 5 in the Appendix. While the device is powered on and running, it remains within a main Active superstate. A separate Alert state can be entered from Active when the sensors detect unsafe metrics, such as a high heart rate or prolonged breathlessness. On entry, the Alert state delivers a high-frequency haptic pattern to prompt the user to consider seeking medical assistance, repeating until the user acknowledges it and the device returns to Active. To acknowledge the alert, the user must deliberately press and hold the buttons for 10 seconds, and the alert event is automatically logged.

Within the Active superstate are four states: Monitoring, User Input, Guided Session and Data Handling. Each comprises further substates with their own transitions, as detailed in Fig. 5. Table I describes each state and substate, while Table II lists every transition with its triggering event and guard condition. The Monitoring state reads sensor data and prompts the user when it determines that a guided breathing session may be suitable. The User Input state handles data logging and self-reported breathlessness, and acts as the interface for beginning a guided session or handling data storage. The Guided Session state delivers a haptic breathing pattern, which is either fixed using a best-guess routine or tailored in real time when the PPG data is considered reliable and motion artefacts are low. Finally, the Data Handling state stores session information and, with the user's consent, exports data to a third-party platform accessible to clinicians.

V. VALIDATION OF FEASIBILITY

Much of the proposed device is already well supported by existing literature. The wristwatch form factor and photoplethysmography (PPG) sensing modality are widely adopted in consumer and research-grade wearables, and heart rate (HR) estimation from wrist PPG is a relatively mature problem, with numerous validated algorithms achieving high accuracy even during motion.

Respiration rate (RR) estimation from wrist PPG, however, remains an active area of research, largely because the respiratory-induced modulations of the PPG waveform are considerably weaker and more susceptible to corruption than the pulsatile component used for HR estimation. There is currently no consensus on the optimal signal processing pipeline for extracting RR from wrist PPG, and reliable RR estimation during motion, such as exercise, is essentially unsolved.

Given that the proposed device relies on RR as a key physiological input for estimating breathlessness, it is necessary to establish whether a reasonable RR estimate can be obtained from wrist PPG under realistic conditions before committing to this sensing approach. To this end, this section presents a

validation analysis using raw PPG data from two randomly selected participants performing sitting and walking activities, drawn from the publicly available PPG-DaLiA dataset [10]. The two participants, referred to throughout by their dataset identifiers S7 and S8, also offer a useful contrast in self-reported baseline fitness: S7 is a 21-year-old female who reported rarely engaging in sport activity, while S8 is a 43-year-old male who reported exercising several times per week. This analysis does not attempt to solve the broader problem of reliable RR estimation from wrist PPG during exercise, but instead aims to validate that a reasonable estimate can be obtained when compared against a ground truth respiration measurement. If reliable RR data can indeed be obtained in this way, then coupled with HR data, it would be recommended to conduct a further study into whether these two parameters alone form a sufficient basis to classify a patient’s breathlessness severity, compared against their own self-assessment. Furthermore, despite this not addressing RR estimation from wrist PPG during motion itself, this is the intended use of the device: gathering this data during rest periods when the wearer is still, such as breaks during sports or rests between weightlifting exercises, to guide the wearer through breathing exercises that aid their recovery.

A. Ground Truth Derivation and the Effect of Motion

For each participant’s sitting and walking sections, a respiration rate ground truth was derived from the chest-mounted RespiBAN unit’s respiration (impedance pneumography) signal, rather than being taken from a single breath-detection algorithm. Three independent breath-detection methods – time-domain peak detection, zero-crossing counting, and spectral dominant-frequency estimation – were each run over 8 s windows (2 s shift) of the band-pass filtered (0.10–0.50 Hz) chest respiration signal, and a window’s rate was only accepted once at least two of the three methods agreed to within 6 breaths/min; disagreeing or low-quality windows were rejected rather than reported. For the sitting sections, shown in Fig. 2, this ensemble produced an accepted rate for 97% (S7) and 89% (S8) of windows, with a smoothly varying respiration rate (mean 19.7 ± 1.4 and 18.9 ± 1.6 breaths/min respectively) consistent with expected resting adult respiration rates – giving confidence that this is a reasonable ground truth to validate the wrist PPG-derived estimate against.

The walking sections were considerably less reliable: the accepted-window rate dropped to 39% (S7) and 46% (S8), and the windows that were still accepted were far noisier (standard deviation rising from ~ 1.5 to ~ 5 breaths/min). Fig. 3 illustrates the source of this degradation at the wrist: the raw wrist PPG signal, which shows a clean, near-periodic pulse train of amplitude ~ 20 – 100 (a.u.) during sitting, is dominated by large, irregular motion artefacts (peaks exceeding ± 400) throughout the walking section, with no consistently identifiable pulse structure. Because the chest strap is more rigidly coupled to the torso than a wristband is to the wrist, its signal degrades less severely under motion, but even this reference becomes considerably less trustworthy while walking. Consistent with

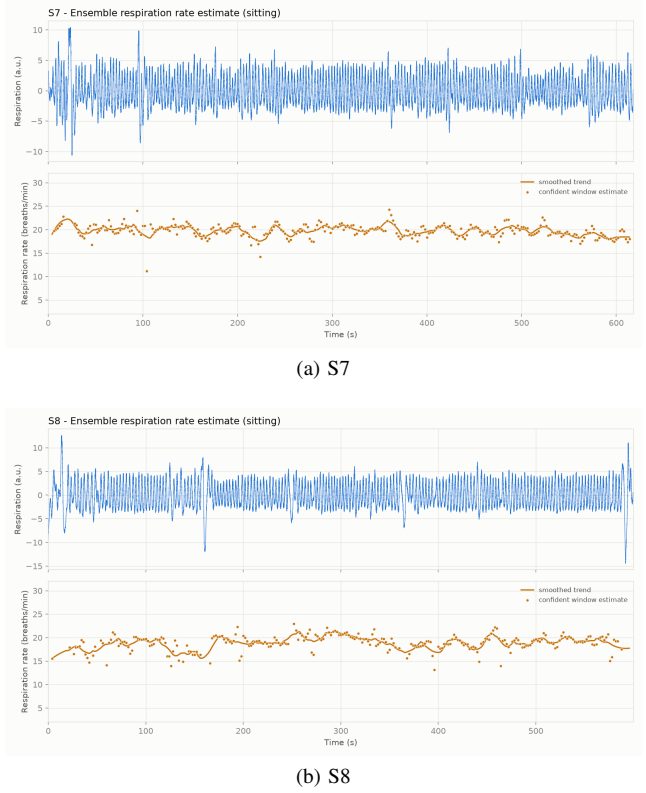


Fig. 2. Ground truth respiration rate estimation for the sitting sections of (a) S7 and (b) S8: raw chest RespiBAN respiration signal (top) and the resulting fused rate at each accepted window, with a smoothed trend (bottom). The high proportion of accepted windows (97% S7, 89% S8) and low estimate variance (± 1.4 – 1.6 breaths/min) support the use of this ensemble as ground truth for validating the wrist-PPG-derived estimate.

the literature discussed above, RR estimation from the wrist PPG during walking was therefore not pursued further in this validation; the remainder of this section focuses on the sitting data, where both the ground truth and the wrist-PPG-derived estimate are reasonably trustworthy.

B. Respiration Rate Estimation from Wrist PPG (Sitting)

For the sitting sections, respiration rate was estimated from the wrist PPG signal alone, with no reference to the chest strap, by exploiting three independent ways in which respiration modulates a PPG pulse train: frequency modulation of the pulse-to-pulse interval (respiratory sinus arrhythmia), amplitude modulation of pulse amplitude (respiration-driven changes in stroke volume), and baseline wander of the PPG’s DC level (respiration-driven changes in peripheral blood volume). Individual heartbeats were first identified from the wrist PPG using an incremental-merge peak/onset detector, with beat-level rejection of implausible pulse intervals or amplitudes. Each of the three respiration-modulated streams was then reconstructed on a common time grid from these detected beats, band-pass filtered to the breathing band (0.10–0.50 Hz), and converted into a breathing rate every 2 s using peak-interval statistics from a rolling 24 s window; the three streams’ rate estimates

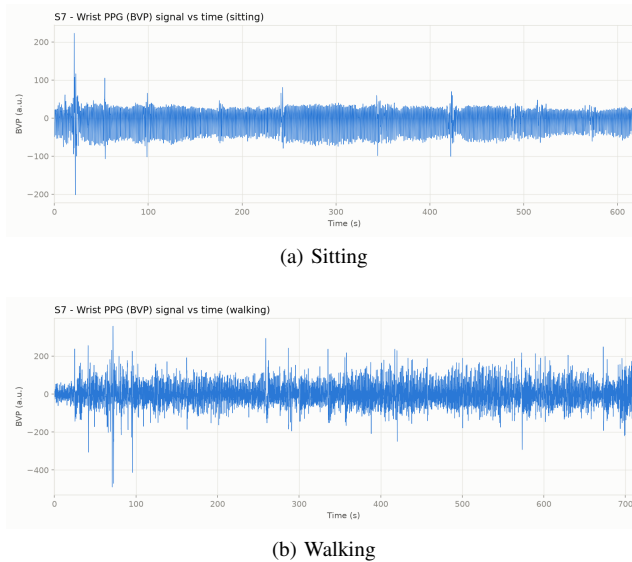


Fig. 3. Raw wrist PPG (BVP) signal for participant S7 during (a) sitting and (b) walking, over the same amplitude-comparable time windows. The near-periodic pulse train visible at rest is replaced by large, irregular motion artefacts once walking, with no consistently identifiable pulse structure – the reason wrist-PPG respiration estimation during motion was not attempted in this validation.

were fused with a quality-weighted vote, discarding a window’s estimate where the streams disagreed excessively.

An initial version of this pipeline reproduced the correct within-window shape of the reference respiration trend but consistently underestimated its level, by around 3 breaths/min for S7 and 4 breaths/min for S8 – a fairly constant per-participant offset rather than one that varied over the session. This is expected: none of the three respiration-modulated streams carries any absolute information about a person’s mean breathing rate, only about its relative fluctuation, so the fused estimate settles at whatever level its own beat-to-beat modulations happen to sit at. The pipeline was therefore revised to include a one-off per-participant DC baseline calibration: a single additive offset (+2.5 breaths/min for S7, +3.6 breaths/min for S8), fitted once against the reference respiration rate over the session, analogous to a short calibration a real device might perform once against a reference measurement before regular use. This step does not alter the underlying detection method or the breath-to-breath shape it recovers – only the constant level it is centred on.

Fig. 6 and Fig. 7, given in the appendix, show the resulting respiration rate for S7 and S8 respectively, compared against the RespiBAN ground truth over the full sitting section. After DC calibration, the mean absolute error against the ground truth fell from 3.1 to 1.8 breaths/min for S7, and from 4.3 to 2.2 breaths/min for S8 (RMSE: 3.8 to 2.3, and 5.0 to 2.8 breaths/min respectively). The estimate does not track every individual fluctuation in the reference trace – window-to-window correlation remains weak – reflecting how subtle the respiratory modulation of a wrist PPG pulse is relative to the noise floor discussed in the introduction to this section, but the overall level and slow trend are well matched throughout both

six- to ten-minute sessions. As a secondary check, heart rate derived from the same wrist PPG beat detector was compared against the chest-strap reference heart rate over the same windows: for S7 the tracking was close (MAE 2.0 bpm), while for S8 it was periodically corrupted by isolated motion artefacts even during nominal sitting, producing brief overestimation spikes (up to 110 bpm against a true rate near 60 bpm) and a correspondingly higher MAE (9.6 bpm). Heart rate estimation is not the focus of this validation, but this nonetheless supports the premise that pulse-level HR detection from wrist PPG is the more mature, lower-risk half of the sensing pipeline relative to RR.

C. Summary

This validation demonstrates that a reasonable respiration rate estimate can be derived from wrist PPG data alone, without any additional sensing modality, provided the wearer is reasonably still. It does not establish whether HR and RR alone are sufficient to predict patient-reported dyspnoea with strong accuracy; a further study, using a validated dyspnoea scale as ground truth, would be needed to determine this. It was also found that motion artefacts severely degrade the accuracy of the wrist-PPG RR estimate, a roadblock already well reported in the literature and consistent with the difficulty of RR estimation from wrist PPG during motion discussed at the outset of this section. The methodology used here is furthermore simple and illustrative rather than conclusive: a larger sample, spanning a range of genders, ages and activity levels across a wider range of activities, should be analysed before drawing firm conclusions, but this is outside the scope of the present report. The key takeaway is that signal processing of the raw wrist PPG signal is the most integral and challenging aspect of realising this product. This validation shows that it is achievable when the wearer is reasonably still – a condition well matched to the product’s intended use, in which the recovery phases of exercise (for example, breaks during sports or rests between weightlifting sets) are used to deliver guided breathing exercises tailored to the wearer’s own derived heart rate and respiration rate.

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APPENDIX

Fig. 4 shows the full electrical schematic of the device, as described in Section IV-B.

Fig. 5 shows the full finite state machine governing the device's operation, as described in Section IV-C.

Table I and Table II detail the states and transitions of the FSM in Fig. 5.

Fig. 6 and Fig. 7 below are referenced from Section V-B, and give the full per-participant wrist-PPG respiration rate results for S7 and S8 respectively.

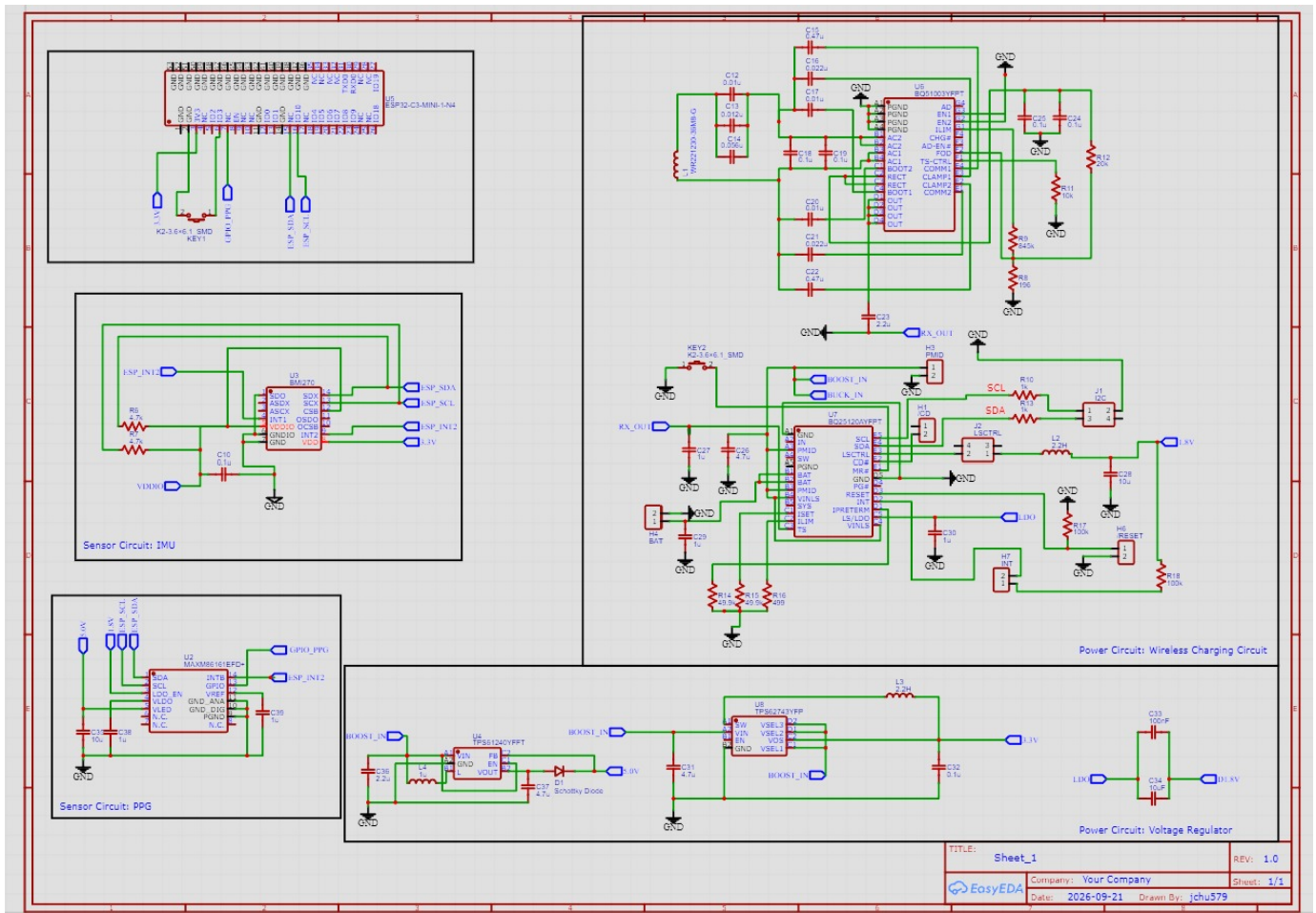


Fig. 4. Electrical schematic of the wearable device, comprising the ESP32-C3 microcontroller, the BMI270 IMU and MAXM86161 PPG sensor circuits, the wireless charging circuit (BQ51003 Qi receiver and BQ25120A charger), and the voltage regulation circuit (TPS61240 boost and TPS62743 buck converters).

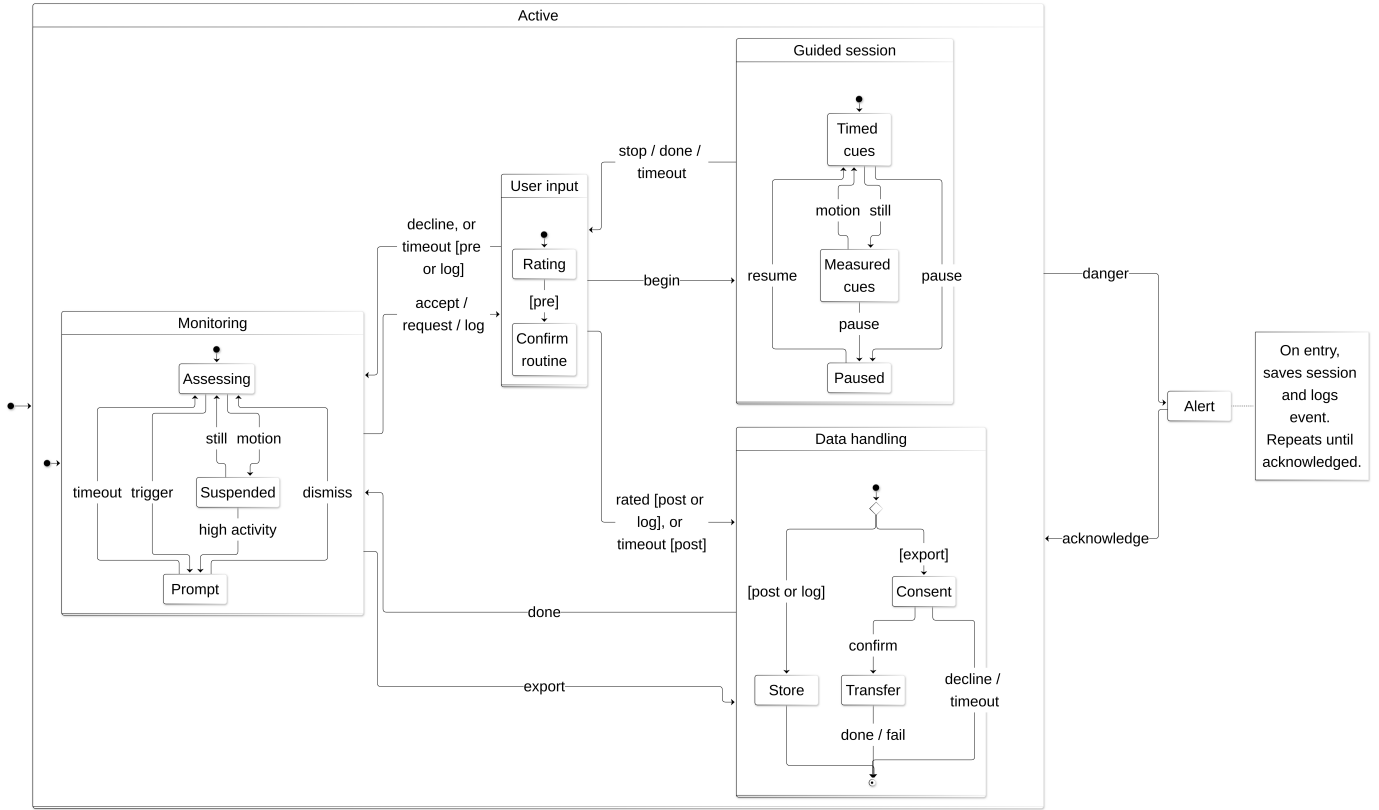


Fig. 5. Finite state machine governing the device's operation. The Active superstate contains the Monitoring, User Input, Guided Session and Data Handling states, each with its own substates and transitions. The separate Alert state is entered from Active when unsafe metrics are detected, and delivers a repeating high-frequency haptic alert until acknowledged.

TABLE I
FSM STATES AND SUBSTATES

State	Sub-state	Description
Monitoring	Assessing	Signals are reliable, so HR, RR and SpO ₂ are estimated and breathlessness is assessed at rest.
	Suspended	Motion makes the PPG signal unreliable, so physiological assessment pauses and only IMU activity is tracked.
	Prompt	A distinct haptic pattern suggests the user rests and begins a breathing exercise.
User input	Rating	The user enters a Borg CR10 score with the buttons; the entering transition sets its context (pre, post or log).
	Confirm routine	The device proposes a breathing routine, which the user accepts or overrules.
Guided session	Timed cues	Inhale, exhale and hold cues follow a fixed schedule without relying on sensor data.
	Measured cues	Cues continue while wrist PPG estimates breathing phases and compares their timing with the cues.
	Paused	Cues are halted at the user's request while the session timer continues.
Data handling	Store	Closes and saves the session recording and any associated ratings.
	Consent	The user must explicitly confirm before any data leaves the device.
	Transfer	The stored data is sent to the clinician.
Alert		A repeating high-frequency haptic pattern advises the user to seek medical help; on entry it saves any open session and logs the event.

TABLE II
FSM TRANSITIONS, WITH TRIGGERING EVENTS AND GUARD CONDITIONS

From	To	Event [guard]	Description
Start	Assessing	power on	Monitoring begins when the device starts up.
Assessing	Suspended	motion	Movement makes the PPG signal unreliable.
Suspended	Assessing	still	The user becomes still and the signal is reliable again.
Assessing	Prompt	trigger	Breathlessness or high HR/RR relative to the resting baseline is detected at rest.
Suspended	Prompt	high activity	The IMU detects prolonged high activity.
Prompt	Assessing	dismiss	The user declines the suggested exercise.
Prompt	Assessing	timeout	The user does not respond to the prompt within the set time.
Monitoring	Rating	accept / request / log	The user accepts a prompt, requests an exercise, or logs symptoms, setting the rating context.
Rating	Confirm routine	[pre]	A pre-session rating leads to routine selection.
Confirm routine	Timed cues	begin	The user confirms the routine and the session starts.
User input	Monitoring	decline, or timeout [pre or log]	The user declines, or does not respond before a session or during logging.
User input	Data handling	rated [post or log], or timeout [post]	A post-session or logged rating is saved, or the session is saved unrated if the user does not respond.
Timed cues	Measured cues	still	The user becomes still, so breathing timing can be measured.
Measured cues	Timed cues	motion	Movement makes measurement unreliable, so cues continue on the fixed schedule.
Timed / Measured	Paused	pause	The user pauses the session.
Paused	Timed cues	resume	The user resumes, and cues restart on the fixed schedule until stillness is confirmed.
Guided session	Rating	stop / done / timeout	The user stops, the routine completes, or the 30-minute cap is reached, leading to a post-session rating.
Monitoring	Data handling	export	The user requests export of stored data.
Data handling	Store	[post or log]	Rating and session data are routed to local storage.
Data handling	Consent	[export]	Export requests are routed to the consent step.
Consent	Transfer	confirm	The user explicitly consents to sending the data.
Consent	Monitoring	decline / timeout	The user refuses or does not respond, and no data is sent.
Transfer	Monitoring	done / fail	The transfer completes or fails, and the device returns to monitoring.
Store	Monitoring	done	Saving completes and the device returns to monitoring.
Active	Alert	danger	Dangerous vitals or a very high rating are detected in any state.
Alert	Assessing	acknowledge	The user acknowledges the alert and monitoring restarts.

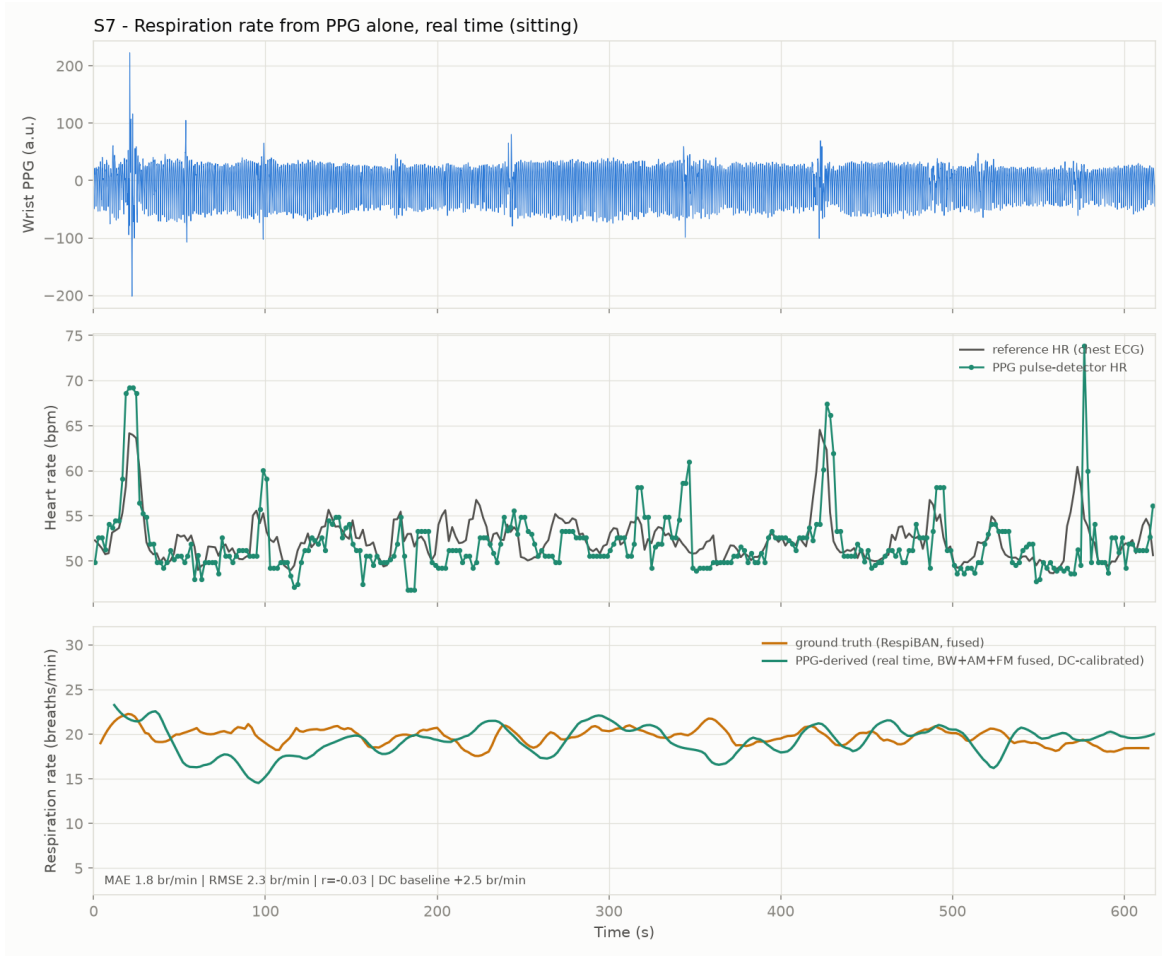


Fig. 6. Participant S7, sitting: wrist PPG (top); heart rate from the chest ECG reference vs. the wrist-PPG pulse detector (middle); and respiration rate from the RespiBAN ensemble ground truth vs. the wrist-PPG-derived estimate, after per-participant DC baseline calibration (bottom). The PPG-derived respiration rate closely tracks the reference level and slow trend (MAE 1.8 breaths/min, RMSE 2.3 breaths/min).

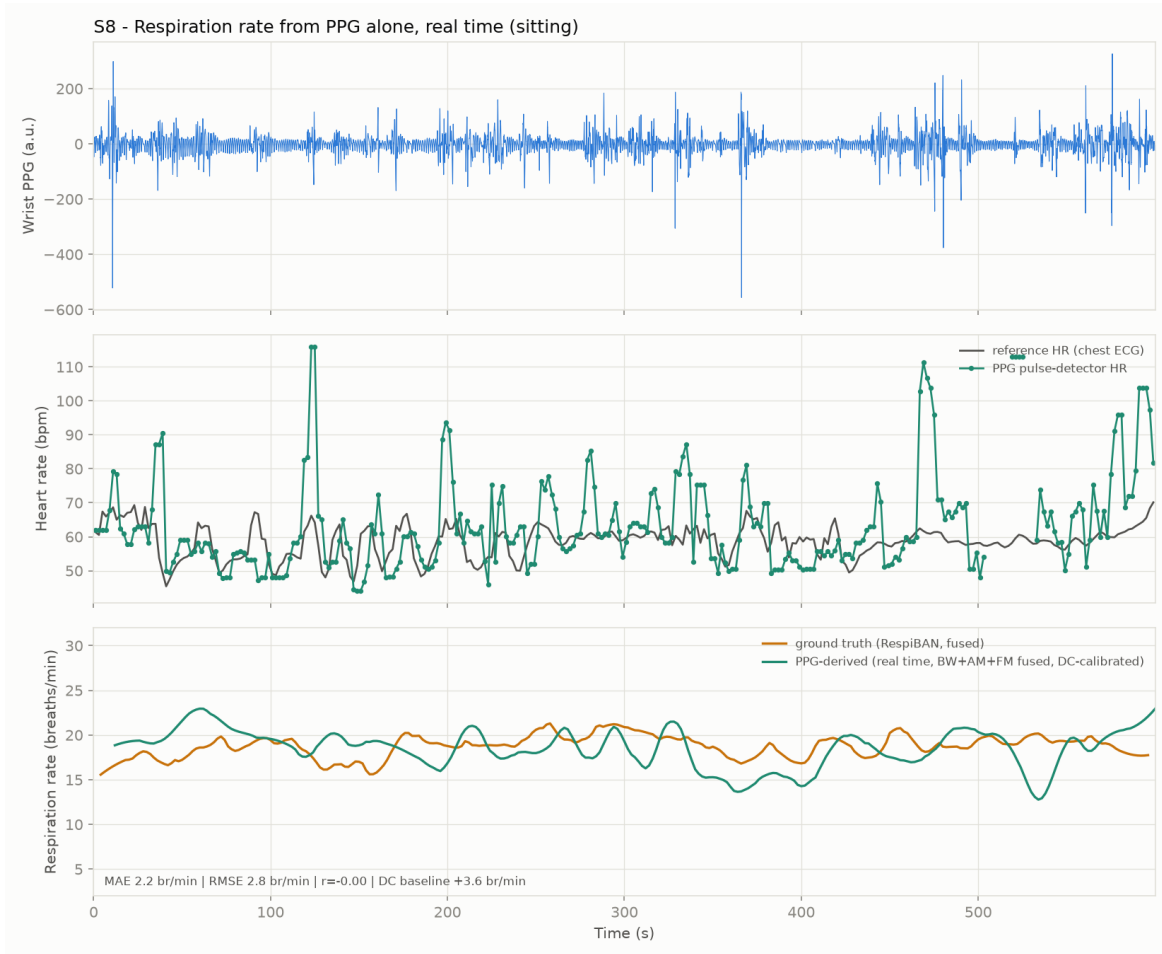


Fig. 7. Participant S8, sitting: wrist PPG (top); heart rate from the chest ECG reference vs. the wrist-PPG pulse detector (middle); and respiration rate from the RespiBAN ensemble ground truth vs. the wrist-PPG-derived estimate, after per-participant DC baseline calibration (bottom). The PPG-derived respiration rate again tracks the reference level and slow trend reasonably closely (MAE 2.2 breaths/min, RMSE 2.8 breaths/min), though heart rate tracking is more affected by intermittent motion artefacts than for S7.